

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

**TECHNOLOGY OF ELASTOMERS
MODEL QUESTION PAPER – SET-1**

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	State the function of PVI with an example.	M 1.04	U
2	List two slow accelerators used in rubber compounding.	M1.02	R
3	List two slow accelerators used in rubber compounding	M2.01	R
4	Name any four polymers that can be cured with Sulphur.	M2.04	R
5	Name two cure meters.	M3.01	R
6	Define moulding.	M3.03	R
7	Amount of process oil required in a recipe for 80phr silica filler is -----	M4.03	U
8	What is the base Hardness of NR?	M4.04	R
9	Enlist two features of EV system.	M2.04	R

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

		Module Outcome	Cognitive level
1	Explain the role of filler in a rubber compound.	M1.03	U
2	Classify antidegradants with one example.	M1.02	U
3	Distinguish gum compound and filled compound.	M2.03	U

4	Mastication is a mechanochemical process. Justify.	M2.03	U
5	State three methods to avoid scorch.	M3.01	U
6	Compare the cure graphs of rubber compound with accelerators DPG and TMTD.	M3.02	U
7	Explain the compounding to meet processing requirement -control of nerve	M4.03	U
8	Discuss the selection of base polymer for abrasion resistant products.	M4.04	U
9	Enlist three advantages of microwave curing.	M3.04	U
10	Calculate the resulting hardness of NR based formulation with 45 phr HAF and 4phr process oil.	M4.03	A

PART C

III. Answer all questions from the following (6 x 7 = 42 Marks)

Module Outcome Cognitive level

1	Illustrate with a flowchart the thermal process for the manufacture of carbon black.	M1.03	U
OR			
2	Explain any four special purpose additives with their functions.	M1.04	U
3	Explain the chemistry of Sulphur vulcanization	M2.04	U
OR			
4	Explain the steps involved in writing a formulation for a rubber product.	M2.02	U
5	Illustrate the cure graphs of rubber compounds with accelerators CBS, TMTM, DPG.	M3.02	U

OR			
6	Describe the continuous vulcanization methods a) Steam tube vulcanization b) Rotocure	M3.04	U
7	Design a rubber compound based on NR with silica as filler for 60 Shore A hardness.	M4.03	A
OR			
8	Calculate the hardness of given formulation. NR-100, ZnO—5 ,Stearic acid -2, HAF-50 ,oil -5 ,PBN –1.0 ,CBS-0.6 , S -2.5.	M4.03	A
9	Describe the determination of structure of Carbon black using DBP absorptiometer.	M1.03	U
OR			
10	Classify the accelerators based on their relative activity with examples..	M1.02	U
11	Design a rubber compound based on SBR using GPF as filler for 65 shore A hardness.	M4.03	A
OR			
12	Describe the factors to be considered while designing a product for maximum tear Resistance.	M4.04	U

Prepared By:

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Lecturer

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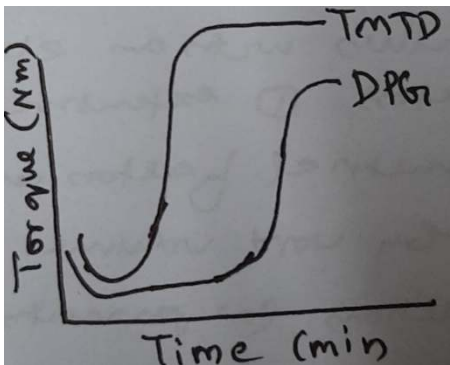
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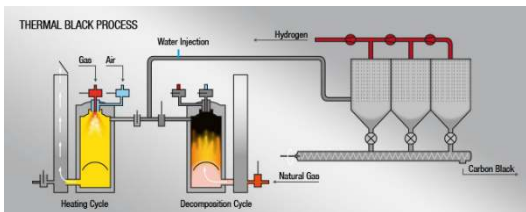
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SCHEME OF VALUATION (SET 1)

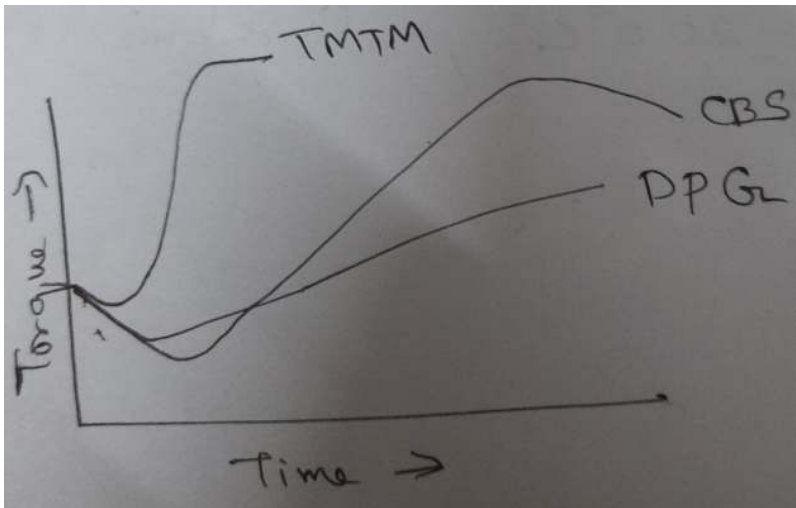
(Scoring Indicators)

Revision : 2021		Course Code :		
Course Title : Technology of Elastomers.				
Qst. No	Scoring Indicator	Split up Score	Sub Total	Total
	PART -A			9
1.	Prevents scorch during processing and storage eg.Salycilic acid		1	
2.	DPG, DOTG.	0.5X2	1	
3	Incorporating compounding ingredients into rubber.		1	
4	NR ,SBR ,PB ,IR.	0.25*4	1	
5	ODR ,MDR	0.5*2	1	
6	Shaping and vulcanisation in a mould at high temp and pressure.		1	
7	8phr		1	
8	38 shore A		1	
9	Good Heat resistance ,poor tensile strength.	0.5*2	1	
	PART – B			
1.	Reinforce ,Extend ,dilute and impart certain processing properties to rubber compound.		3	
2.	Antioxidants — <u>Staining type</u> eg. Amines - PBN - <u>Nonstaining type</u> eg. Phenolics –SP, Antiozonants — <u>Chemical protectant</u> eg. N-isopropyl –N!-phenyl p-phenylenediamine. - <u>Physical protectant</u> eg. Microcrystalline Blended petroleum waxes.	1.5	3	
	Gum compound –rubber compound with filler Filled compound -rubber compound with filler.	1.5		24
3	Mechanically -breakdown of polymer chain,reduction of viscosity in a mixing mill .chemically—chain scission in the	1.5*2	3	

4	polymer chain and reaction with oxygen of atmosphere. Using primary and secondary accelerators Adding retarders Delayed action accelerators are used	1.5*2	3	
5		1*3	3	
6		3*1	3	
7	Control of nerve • Give suitable degree of mastication • Increased filler loadings • Choice of filler loadings • Addition of factice • Addition of partially Xd polymers.			
8.	NR ,SBR for mild abrasion resistance NR/PB and SBR/PB for good abrasion resistance Low particle size blacks and good antioxidant protection Has good influence on abrasion resistance. Able to cure thick products.	1.5 1.5	3	

9.	Reduced energy usage Reduced tooling cost	1*3	3	
10	Hardness of base polymer = 38shore A 1.9 phr HAF - 1shore A hardness So 45 phr HAF -24 shore A hardness 4phr oil – decreases 2 shore A hardness So resulting hardness of formulation =38+24-2= 60 Shore A.	1*3	3	
1.	PART--- C  Raw material storage ---→c-black production and storage --→ C-black separation and cooling --→ C-black pelletising ---→ C-black storage. Explanation OR	1	3	
2.	Abrasives to produce grinding wheels for abrasive property Blowing agents-for producing cellular rubbers. Flame retardants – for flame resistance Internal lubricants – for good mould release.	3	7	
		2	7	
		2		

3.	The diagram illustrates the vulcanization process. At the top, under the label 'Unvulcanized', several wavy lines represent individual rubber molecules. An arrow labeled 'Sulfur' points down to the bottom part of the diagram. The bottom part shows a 'Vulcanized Network' where the rubber molecules are interconnected by 'Crosslinks' represented by black dots. A chemical structure shows two wavy lines connected by a bridge labeled S_x. Vulcanization is a chemical process that converts natural rubber and other polydiene elastomers into cross-linked polymers. The most common vulcanization agent is sulfur. It forms bridges between individual polymer molecules when heated with rubber. ... The cross-linked elastomers have much improved mechanical properties. Sulfur crosslinks are present as monosulphidic, disulphidic, polysulphidic, pendant sulphides, cyclic monosulphides and disulphides. OR • Base polymer selection -depends on service requirements • Selection of vulcanising agent based on base polymer, heat resistance, Tensile strength required for the product. • Selection of accelerator based on curetime, vulcanising agent and accelerator. • Activator selected based on base polymer . • Filler selected based on hardness and modulus required for the product. • Plasticiser selection based on type of base polymer. • Depending on service reqt. A/Os are selected.	2 1	7	
4.		3	4	7

	To get special properties special purpose ingredients are selected.	1*7		
5.	 OR Steam tube vulcanisation for cables.---the cable after extrusion passes through a long steam filled tube sealed at one end to retain steam pressure. This is a continuous vulcanisation method .		7	
6.	Rotocure for beltings The Rotocure in the set-up and consists of a large, steam-heated, revolving steel drum. This is partly encompassed by an endless steel band. The steel band encompasses approximately 2/3 of the circumference of the drum through the action of two conducting rollers and a tensioning pulley. The drum is heated by means of steam to a temperature of 150 – 200°C. In addition, there is a heating plate bent round approximately half of the surface of the drum and which gives some additional heat, but primarily it prevents loss of heat.		7	
7.	Base hardness of NR -38 shore A Difference in hardness =22 shore A 2phr silica – 1 shore A rise in hardness	3		

	Amount of filler required= $22 \times 2 = 44$ phr. Oil required = 6.6 phr Decrease in hardness due to oil = 3.3 shore A Silica filler needed to compensate = $2 \times 3.3 = 6.6$ phr So total silica filler to be added = $44 + 6.6 = 50.6$ phr OR Hardness of base polymer = 38 shore A	4	7	
8.	1.9 phr HAF - 1 shore A hardness So 50 phr HAF - 25 shore A hardness 5 phr oil - decreases 2.5 shore A hardness So resulting hardness of formulation = $38 + 25 - 2.5 = 60.5$ Shore A.	1 1 2		
9.	Structure of Carbon black --- 20 g c-black in a chamber ,DBP is added ,when all voids are filled ,machine get stuck and reading taken. DBP number = Reading X 5. OR	2 1 1	7	
10.	Relative activity of accelerators Slow 90-120 min Aniline Moderately fast 60 min DPG Fast 30 min MBT, MBTS Ultra few mins Thiurams, Dithiocarbamates	2 2 1 1	7	
11.	Base hardness of NR = 38 shore A Difference in hardness = $65 - 38 = 27$ shore A 2.5 phr GPF = 1 shore A rise in hardness Amount of filler required = $27 \times 2.5 = 68$ phr.	2 2 2 1	7	

12.	Oil required =6.8 phr			
	Decrease in hardness due to oil=3.4 shore A	1		
	Silica filler needed to compensate=2 .5X 3.4=8.5 phr	2		
	So total silica filler to be added = 68+8.5 = 76.5 phr	1	7	
		2		
	OR			
	Maximum tear resistance			
	Base polymer -high gum strength polymers --- NR ,CR selected.			
	Fillers – mineral fillers like Al.Silicate reinforcing silica ,fine clay	1		
	CI resin, Petroleum resin ,Phenolic resin improve tear by 5%.			
		5		
		2	7	
	XXXXXXXXX			

Course: Technology of Elastomers

BLUE PRINT -TOE -1

Mark Distribution

Module	Hr/ Module	Marks/ Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks
1	15	32	2	2	2	6	4	28	8	36
2	10	21	3	3	3	9	2	14	8	26
3	15	32	2	2	2	6	2	14	6	22
4	18	38	2	2	3	9	4	28	9	39
Total	58	123	9	9	10	30	12	84	31	123

BLUE PRINT
COGNITIVE LEVEL MARK DISTRIBUTION

Cognitive Level	Marks	% of Marks
Remembering	7	6
Understanding	81	66
Applying	35	28
Analysing		
Evaluating		
Creating		
Total	123	100

Question Wise Analysis – TOE No. 1

Course : Technology of Elastomers .

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M 1.04	Understanding	1	2
I.2	M1.02	Remembering	1	2
1.3	M2.01	Remembering	1	2
I.4	M2.04	Remembering	1	2
1.5	M3.01	Remembering	1	2
1.6	M3.03	Remembering	1	2
1.7	M4.04	Applying	1	2
1.8	M4.04	Remembering	1	2
1.9	M2.04	Remembering	1	2
II.1	M1.03	Understanding	3	7
II.2	M1.02	Understanding	3	7
II.3	M2.03	Understanding	3	7
II.4	M2.03	Understanding	3	7
II.5	M3.01	Understanding	3	7
II.6	M3.02	Applying	3	7
II.7	M4.03	Understanding	3	7

II.8	M4.04	Understanding	3	7
II.9	M3.04	Understanding	3	7
II.10	M4.04	Applying	3	7
III.	M1.03	Understanding	7	17
IV	M1.04	Understanding	7	17
V.	M2.04	Understanding	7	17
VI	M2.02	Understanding	7	17
VII	M3.02	Applying	7	17
VIII	M3.04	Understanding	7	17
IX	M4.02	Applying	7	17
X	M4.04	Applying	7	17
XI	M1.03	Understanding	7	17
XII	M1.02	Understanding	7	17
XIII	M4.02	Applying	7	17
IV	M4.04	Understanding	7	17

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

TECHNOLOGY OF ELASTOMERS
MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A

IV. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Classify fillers according to reinforcement.	M 1.03	U
2	What are physical protectants?	M1.02	R
3	List two general purpose rubbers.	M2.01	R
4	Define compounding.	M2.01	R
5	What is cure rate index?	M3.01	R
6	----- and ----- are used for vulcanization in peachey process.	M3.04	R
7	Parts of HAF filler to increase I shore a hardness in NR is --- -----	M4.02	R
8	As particle size of filler increases hardness -----	M4.01	U
9	Base hardness of NR is -----shore A.	M4.01	R

PART B

V. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

		Module Outcome	Cognitive level
1	Differentiate plasticizer, Softener and extender.	M1.02	U
2	Explain the nomenclature of carbon black.	M1.03	U

3	State 3 importance of base polymer in compound design.	M2.01	U
4	Justify the curing of SBR .	M2.04	U
5	Illustrate the cure graph of a mooney viscometer.	M3.01	U
6	Explain the essential features of autoclave curing..	M3.04	U
7	Find the resulting hardness of an NR based formulation with 50 phr GPF and 5 phr oil.	M4.02	A
8	Find the resulting hardness of SBR based formulation with 65 phr HAF and 6.5 phr oil.	M4.02	A
9	Enlist three advantages of microwave curing.	M3.04	U
10	10.Explain the importance of antidegradants in a rubber compound.	M1.02	U

PART C

VI. Answer all questions from the following (6 x 7 = 42 Marks)

Module Outcome Cognitive level

1	Illustrate with a flowchart thefurnace process for the manufacture of carbon black.	M1.03	U
OR			
2	Write a typical recipe of any rubber product and mention the function of each ingredient.	M1.02	U
3	Explain the changes that happen in a rubber compound after vulcanization.	M2.04	U
OR			
4	Distinguish CV, EV and Semi EV system	M2.04	U
5	Illustrate the cure graphs of rubber compounds with accelerators ZDC, DCBS, DOTG.	M3.02	U

OR			
6	Describe the batch vulcanization methods a)Oven curingb) Peachey process c) Gas curing	M3.04	U
7	Design a rubber compound based on SBR with HAF as filler for 65 Shore A hardness.	M4.02	A
OR			
8	Calculate the hardness of given formulation. NR-100, ZnO—5 ,Stearic acid -2, ISAF-60 ,oil -6 ,PBN –1.0 ,CBS-0.6 , S -2.5.	M4.04	A
9	Describe the procedure to determine the iodine number of Carbon black.	M1.03	U
OR			
10	Define blowing agent and explain its classification with two examples.	M1.02	U
11	Design a rubber compound based on NR using Whiting as filler for 65 shore A hardness.	M4.02	A
OR			
12	Describe the factors to be considered while designing a product for maximum tensile properties.	M4.04	U

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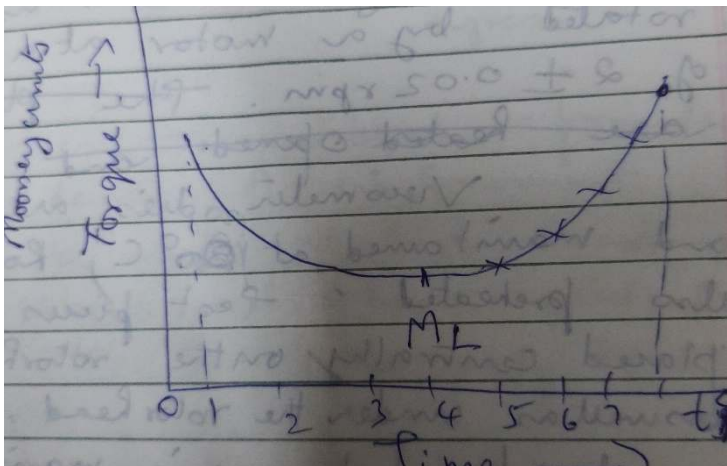
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SCHEME OF VALUATION (SET II)
(Scoring Indicators)

Revision: 2021 Course Code: Course Title: Technology of Elastomers. -2				
Qst. No	Scoring Indicator	Split up Score	Sub Total	Total
	PART -A			9
1.	Reinforcing ,semi reinforcing ,inert	0.5*2=1	1	
2.	Antiozonants for stationary applications.	1	1	
3	NR ,SBR		1	
4	Process of incorporating compounding ingredients into rubber.	1	1	
5	Measure of rate of vulcanisation based on optimum vulcn and incipient scorch time	1	1	
6	SO ₂ and H ₂ S	0.5*2	1	
7	1.9 phr	1	1	
8	Decreases	1	1	
9	38 Shore A	1	1	
	PART – B			
1	Plasticisers -added to rubber compounds to aid the processing operations, added with fillers to reduce compound cost.	1		
	Softeners -Petroleum oils, petroleum jelly when added in high amounts -5 to 10 phr.-reduce viscosity, ease of filler incorporation.	1	3	
	Extenders -when used to extend or reduce the cost of rubber compounds .	1		
	N110 – naming ,One letter followed by two numerals.			24

2.	N indicates normal cure ,S for slow cure. First number indicates particle size or surface area. Second number and third numeral are selected arbitrarily. If std. level of structure second numeral is repeated and the last is zero. Eg.N330 ,N326	3	3	
3	Most important component of a recipe Depends on service requirements Blends /single polymers are used.	1*3	3	
4	SBR is slower curing when compared to NR. Sulfur required is less than NR due to less unsaturation	1.5*2	3	
5		3	3	
6	Autoclave curing Large number of components can be vulcanised at one time Products formed on mandrels and sheetings can be vulcanised.	3*1	3	

7.	Resulting hardness = $38 + 20 - 2.5 = 55.5$ shore A		3	
8.	Resulting hardness = $40 + 34.32 - 3.25 = 70.95$ shore A.		3	
9.	Able to cure thick products. Reduced energy usage Reduced tooling cost	1*3	3	
10	Antidegradants protect the rubber products from external agents like oxidation ,UV rays ,gama rays ,heat light etc. thus prolonging the service life of the product.		3	
PART -C				
1.	FURNACE PROCESS - this process consists of incomplete combustion of natural gas or heavy aromatic residue oils from petroleum/ coal tar industries in refractory lined steel furnaces. – followed by separation from combustion gases - dry or wet pelletisation and storage.	3		
			7	

Flow chart for furnace process for c-black manufacture.

```
graph LR
    Raw[Raw material Storage] --> Furnace[Furnace]
    Furnace --> Cooling[C-black Production and cooling]
    Cooling --> Separation[C-black separation and cooling]
    Separation --> Storage[C-BLACK STORAGE]
    Furnace --> Pelletising[Pelletising]
    Pelletising --> Storage
    Storage --> Storage
```

OR

A General Tyre tread recipe or Formulation

Ingredients	Quantity in PHR	Function-Justification
NR	100	Elastomer
Oil soluble sulfonic acid	2.0	Processing aid
ZnO	5	Activator
Stearic acid	2	Accelerator activator
A/O PBN	2	Antioxidant
Substituted p-Phenylenediamine	4.0	Antiozonant
MC wax	1	Processing aid and finish
Mixed process oil	5	softener
HAF black	50	Filler
Sulphur	2.5	Vulcanisingagent
CBS	0.6	Accelerator
N-nitroso diphenylamine	0.5	Retarder

Crosslinking of polymer chains with vulcanising agent to form three dimensional structures. -elasticity increases

Rubber loses its tackiness.

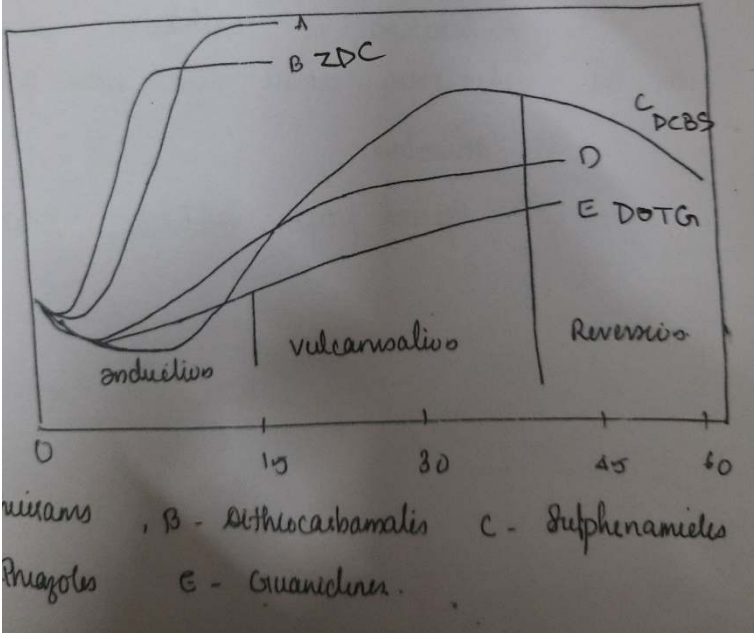
Become insoluble in solvents

Ageing resistance increases.

Overall vulcanizate property increases

OR

CV	EV	SEV
More S	Less S	in-between

	Good tensile strength	Poor tensile strength	Moderate values	2		
	Poor heat resistance	Excellent heat resistance	Moderate properties.	2		
5	 Induction Vulcanisation Reversion A Thiocarbamates B Dithiocarbamates C Sulphenamides D Phthalates E Guanidines			3		
6.	OR Oven curing -for certain heat resistant polymers -silicone and fluorocarbon rubbers post curing done with oven cure. Peachy process – cold curing with SO₂ followed by H₂S gas eg .rainwear ,proofings Gas curing -Done using a jacketed autoclave -jacket filled with high pressure steam ,while interior filled with high pressure gas Nitrogen or coal gas			3 2 2	7	
7.	Base hardness of NR -40 shore A			2		

	Difference in hardness = 25 shore A 2.3 phr HAF – 1 shore A rise in hardness Amount of filler required = $25 \times 2.3 = 57.5$ phr. Oil required = 5.75 phr Decrease in hardness due to oil = 2.7 shore A Silica filler needed to compensate = $2.7 \times 2.3 = 6.21$ phr So total silica filler to be added = $57.5 + 6.2 = 63.7$ phr OR	2		
8	Resulting hardness = $38 + 35.3 - 3 = 70.3$ phr Explain with the steps.	3		
9	To 1 g carbon black add 50 ml iodine solution and shaken, centrifuged. The excess iodine is titrated with std. thiosulphate solution and adsorbed iodine is expressed as mg per g of c-black adsorbed. Iodine no. = $B - S/B \times VN/W \times 126.91$ OR	2		
10	Blowing agent organic eg Ammonium bicarbonate, NaHCO_3 Inorganic eg. DNPT, ADC Definition A blowing agent is a substance which is capable of producing a cellular structure via a foaming process in a variety of materials that undergo hardening or phase transition, such as polymers, plastics, and metals. They are typically applied when the blown material is in a liquid stage.	2	7	

11	Hardness of base polymer = 38shore A 6.4 phr whiting - 1 shore A hardness So 50 phr HAF -25 shore A hardness 5 phr oil – decreases 2.5 shore A hardness So resulting hardness of formulation = $38+25-2.5= 60.5$ Shore A.	4 3		
12.	Tensile strengths are obtained using high gum strength polymers like NR,IR ,CR and isoprene acrylonitrile copolymer. Low gum strength rubbers require fine particle size blacks .low particle size blacks or reinforcing silicas ,30-60 phr range. HSR resins produce increased strength levels in SBR ,Crosslinkable phenolic resins in NBR etc., &&&&&&&&&&&&&&	2 3 2		7

Course: Technology of Elastomers

BLUE PRINT -TOE -2

Mark Distribution

Module	Hr/ Module	Marks/ Module ($h_i / \Sigma H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks
1	15	32	2	2	3	9	4	28	9	39
2	10	21	2	2	2	6	2	14	6	22
3	15	32	2	2	3	9	2	14	7	25
4	18	38	3	3	2	6	4	28	9	37
Total	58	123	9	9	10	30	12	84	31	123

BLUE PRINT
COGNITIVE LEVEL MARK DISTRIBUTION

Cognitive Level	Marks	% of Marks
Remembering	7	6
Understanding	72	58
Applying	44	36
Analysing		
Evaluating		
Creating		
Total	123	100

Question Wise Analysis – TOE No. 2

Course : Technology of Elastomers .

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M 1.03	Understanding	1	2
I.2	M1.02	Remembering	1	2
1.3	M2.01	Remembering	1	2
I.4	M2.01	Remembering	1	2
1.5	M3.01	Remembering	1	2
1.6	M3.04	Remembering	1	2
1.7	M4.02	Remembering	1	2
1.8	M4.01	Understanding	1	2
1.9	M4.01	Remembering	1	2
II.1	M1.02	Understanding	3	7
II.2	M1.03	Understanding	3	7
II.3	M2.01	Understanding	3	7
II.4	M2.04	Understanding	3	7
II.5	M3.01	Understanding	3	7
II.6	M3.04	Understanding	3	7
II.7	M4.02	Applying	3	7

II.8	M4.02	Applying	3	7
II.9	M3.04	Understanding	3	7
II.10	M1.02	Understanding	3	7
III.	M1.03	Understanding	7	17
IV	M1.02	Understanding	7	17
V.	M2.04	Understanding	7	17
VI	M2.04	Understanding	7	17
VII	M3.02	Applying	7	17
VIII	M3.04	Understanding	7	17
IX	M4.02	Applying	7	17
X	M4.04	Applying	7	17
XI	M1.03	Understanding	7	17
XII	M1.02	Understanding	7	17
XIII	M4.02	Applying	7	17
IV	M4.04	Understanding	7	17